

The Mountain's Instructions

Populations that live where oxygen is thin have already run the experiment that oxygen-carrier designers keep trying to repeat, and they arrived at answers that look strange from a hospital

Nobody designs a blood substitute by looking at a Himalayan village, and that is the first mistake. Every chemist working on hemoglobin-based products carries an assumption so deep that it rarely gets stated: more oxygen-carrying capacity is better. If a patient is anemic, add hemoglobin. If tissue is starved, add hemoglobin. If the first product fails, make a more concentrated one. The assumption feels like arithmetic.

Populations who have lived for thousands of years at four thousand meters, and animals that dive for an hour on a single breath, have been running a much longer experiment on that assumption, and their answer is more complicated. In several of them, evolution pushed hemoglobin concentration down, not up. In others it pushed the storage of oxygen out of the blood altogether. Reading their results carefully does not give a designer a blueprint. It gives a set of warnings and a set of questions, and I think that is the more useful thing.

Two mountains, two answers

Consider two large human populations who have lived at high altitude for a long time. The Tibetan plateau sits above four thousand meters, and the high Andean altiplano is similar. The air at those altitudes contains the same fraction of oxygen as at sea level, but the pressure is roughly forty percent lower, so each breath delivers correspondingly less. Every resident faces the same physical problem.

Their solutions differ. Andean highlanders tend to respond the way lowlanders do when they climb: their bodies make more red cells, and hemoglobin concentration rises, in some individuals to very high values. This is an adaptation of a kind, since more hemoglobin means more oxygen per liter of blood. Tibetans do something else. A review of the research on hemoglobin at altitude summarizes their pattern as a relatively low erythropoietic response with a correspondingly low hemoglobin concentration. Tibetans living on the plateau have hemoglobin values that are strikingly close to those of people at low elevation, or only modestly higher.

If you had to bet before looking at the data, you would have bet on the Andean strategy. It is the one a physician would recommend. The puzzle is that the Tibetan strategy appears to work better.

The price of thick blood

The explanation begins with a property of blood that clinicians know but rarely feel: viscosity. Blood is a suspension of cells, and as the fraction of cells rises, the fluid thickens, slowly at first and then steeply. Thicker blood is harder to pump, flows more sluggishly through small vessels, and can reduce delivery to tissues even though it contains more oxygen per milliliter.

The relation between cell fraction and oxygen delivery therefore has a peak. Below the peak, adding cells helps, because capacity dominates. Above it, adding cells hurts, because viscosity dominates. Physiologists have estimated where the optimum falls in various circumstances, and it is lower than most people expect, and it moves with conditions.

The Andean pattern can overshoot. Chronic mountain sickness is a syndrome of excessive red cell production at altitude, and it brings headaches, fatigue, blue lips, and an increased risk of stroke and heart attack. The review I mentioned reports that among Peruvian Andean males the condition can reach rates as high as 18 percent, while among Tibetans it is around 1 percent. In other words, the population that made the intuitively sensible response paid for it in disease, and the population that held back did not.

This is not a story about one group being better than the other. Genetic and developmental factors differ, and Andeans have other adaptations, including changes in blood vessel function. It is a story about the shape of a curve: there is a range where more is better, and beyond it there is a range where more is worse. That shape is the first instruction the mountain gives.

The gene that turned the dial down

The molecular story is unusually clear for a human adaptation. A gene called EPAS1 encodes a protein known as HIF-2 alpha, a transcription factor that helps cells respond to low oxygen, and among its targets is the erythropoietin gene, the master regulator of red cell production. Tibetans carry variants of EPAS1 that appear to reduce its activity. The review states that these variants function as loss-of-function alleles that downregulate the protein product, and that they reach a frequency of roughly 72 percent in Tibetan populations.

The origin of the variant was a surprise. In 2014 researchers reported that the Tibetan version of the gene sits within a stretch of DNA that closely resembles the genome of Denisovans, an extinct group of ancient humans known from a cave in Siberia. The finding suggests that a variant useful for surviving on the plateau was inherited through interbreeding with an archaic population and then favored by natural selection in the mountains.

The ironic part is worth stating clearly. A gene variant that reduces the body's response to low oxygen was selected for in people living where oxygen is scarce. That sounds paradoxical until you recognize that the body's response to low oxygen is a blunt instrument. Ramping up red cell production in response to every dip in oxygen level is a strategy that suits a short trip to altitude, and can be counterproductive over a lifetime.

What else the Tibetan body does

If Tibetans do not compensate by adding hemoglobin, they must compensate some other way, and the research points to several. Their resting ventilation tends to be high, they show relatively strong oxygen delivery through increased blood flow to certain tissues, and studies have reported higher levels of nitric oxide in their circulation than in lowlanders at the same altitude, which would dilate vessels and improve flow. Some evidence suggests differences in how they use oxygen within cells.

I would not lean too hard on any single item on that list; the field is active and the picture is still being filled in. What the collective evidence suggests is that the Tibetan strategy favors

delivery through flow and efficiency rather than through capacity. It keeps blood thin, keeps vessels open, and makes each molecule of oxygen count.

For a designer of oxygen carriers, this is instructive. The tissue does not need more hemoglobin. It needs oxygen delivered at the right place, at the right rate, through vessels that stay open. Those are three separate targets, and adding hemoglobin addresses only the first.

The diving animals

Move now from mountains to the sea. A Weddell seal can dive for tens of minutes on one breath, and some whales can dive for far longer, while a human freediver, with training, can reach several minutes. The seals do not have magical lungs. In fact, diving mammals often carry relatively little oxygen in their lungs, because compressed air in the chest is a hazard at depth. They carry it elsewhere.

They store oxygen in three places. First, they have unusually large blood volume relative to body size. Second, their blood is rich in hemoglobin, and in some seals a contracting spleen squeezes a reservoir of packed red cells into the circulation during a dive, raising the concentration temporarily, a mechanism that lets the animal hold thin blood at rest and thick blood when it is needed. Third, and most strikingly, their muscles contain very high concentrations of myoglobin, the single-chain relative of hemoglobin that stores oxygen inside muscle cells. Seal and whale muscle is nearly black from it.

The design principle is different from that of the mountain dweller. Here the animal maximizes stored oxygen, and it does so by putting the storage where it is used, inside the muscle, so that the oxygen does not have to travel. The blood serves as a supplemental reservoir and a transport line, but the emergency supply sits at the point of need.

Why storage matters to a carrier designer

Storage and transport are usually confused in discussions of oxygen carriers, and the confusion has practical consequences.

A transport system moves oxygen from where it is plentiful to where it is needed. Its value depends on flow. If flow stops, the transport system delivers nothing, however full its carriers.

A storage system holds oxygen where it is needed. Its value depends on how much it holds and how fast the tissue draws on it. If flow stops, the store lasts until it is empty.

Human blood evolved as a transport system with a modest reserve. The venous blood returning to the lungs is still about three-quarters saturated at rest, which is a reserve, but not a large one. When flow stops, as in a cardiac arrest or a clamped vessel, the reserve is spent within minutes.

A cell-free carrier is also a transport system, and its main potential contribution is to reach places that red cells cannot. But it is worth asking whether there are situations in which a carrier could act more like storage. An organ held cold on a machine is one. If the preservation fluid holds a high dissolved oxygen content in the form of a carrier, the tissue has a small reserve to draw on even without flow. A tissue starved by a blockage, briefly, could use the same reserve. I raise this not as a claim about existing products but as a conceptual tool: when

you evaluate a carrier for a particular use, ask whether the job is transport or storage, because the answer changes what properties matter.

Affinity, revisited from the rooftop of the world

Another lesson from altitude concerns oxygen affinity, the tightness with which hemoglobin grips oxygen. In another post I described how the P50 is the pressure at which hemoglobin is half saturated, and how the tissue's needs favor different values in different jobs.

Animals at altitude show the effect in a beautiful way. The bar-headed goose migrates across the Himalayas at altitudes that would incapacitate most birds, and its hemoglobin is reported to have a higher oxygen affinity than that of lowland geese, a difference traced to changes in the amino acid sequence. Llamas, adapted to the Andes, also carry hemoglobin with higher affinity than other mammals of their size. The logic is plain: at low pressure, a hemoglobin that grips harder loads more completely in the lung, and the tissue still receives what it needs because the pressure drop from lung to tissue is steep.

Humans at altitude do something related. A shift in the oxygen-binding curve toward higher affinity is seen in some highland populations and in pregnant women, and in the fetus, whose hemoglobin has a higher affinity than an adult's so that it can extract oxygen across the placenta.

The design implication runs against the intuitions of a first-time designer. The best affinity is not fixed; it depends on the environment. A product intended for a patient breathing thin air in an ambulance at altitude, or a patient with a lung problem that impairs loading, might want a higher affinity than one intended for a patient with a normal lung and a starved limb. There is no single right P50. The mountain tells you that the answer depends on where you live.

The lesson about dilution

There is a further link, and it involves a therapy that emergency physicians know well: hemodilution. When a patient loses blood and receives fluid, the remaining red cells are diluted. The patient's hematocrit falls, and blood becomes thinner. It is easy to think of this as pure harm, and if it goes too far, it is.

But moderate dilution improves flow through small vessels by reducing viscosity, and the body compensates for the lower capacity by pumping more blood. There is a range of hematocrit, lower than normal, in which delivery is about as good as or better than at normal values. The Tibetan pattern, seen from this angle, is a permanent version of controlled hemodilution.

An oxygen carrier dissolved in plasma is a way to have it both ways. It adds oxygen capacity without adding cells, so it raises capacity while leaving viscosity low. In principle, it lets a patient be thin-blooded and well-supplied. This is the deepest of the arguments for the whole class of products, and the mountain reinforces it. The corresponding risk is that the dissolved carrier brings its own problems, discussed elsewhere on this site, including the scavenging of nitric oxide, which is exactly the molecule that the Tibetan body appears to keep at higher levels. The highlander's physiology and the dissolved carrier pull in opposite directions on that one variable.

A worked example of the peak

A small calculation makes the shape of the curve concrete. The numbers are illustrative simplifications and should not be read as physiological constants.

Suppose oxygen delivery is proportional to two things multiplied together: the oxygen content of a liter of blood, which rises in direct proportion to the fraction of cells, and the flow rate the heart can sustain, which falls as viscosity rises. Take a hematocrit of forty-five percent as the baseline, with an oxygen content of 1.0 in arbitrary units and a flow of 1.0. At a hematocrit of thirty, the content falls to about 0.67, but the flow rises, because thinner blood moves more easily. If the flow improves by, say, thirty percent, the product is 0.87, a modest loss. At sixty percent, the content rises to 1.33, but if flow drops by forty percent because the blood has thickened badly, the product is 0.80. Both extremes are worse than the middle.

The exact peak depends on the vessel, the heart and the person, so this is only an outline. But it shows why a physician who sees a low hemoglobin should not automatically see a problem, and why a physician who sees a very high hemoglobin should not automatically see an advantage.

The athlete's shortcut

The sport of endurance racing supplies a second natural experiment, and an uncomfortable one. Athletes have long known that training at altitude raises red cell mass, and some have used shortcuts, including transfusion of their own stored blood and injections of erythropoietin, to raise hematocrit further. These practices are banned in sport for good reasons, including fairness and safety.

The safety story is a direct illustration of the viscosity problem. Endurance athletes who pushed their hematocrit beyond natural limits were reported in the 1990s to have suffered serious cardiovascular events, and the sport introduced hematocrit limits, initially as a health measure. The lesson is the same as the mountain's: raising capacity by making blood thicker has a ceiling, and exceeding it can be lethal.

The athlete's case also points at a design opportunity. If capacity can be raised without thickening the blood, the ceiling moves. That is precisely what a dissolved carrier promises, and precisely why the sporting world has watched such products with anxiety. Some carriers have appeared on lists of prohibited substances, as they were used or considered for use in sport, and their detection is a live topic in anti-doping laboratories. It is an unexpected reminder that the same property that might help a bleeding patient could distort a race.

Cautions from the mountain

It would be irresponsible to end on the note that nature has shown the way. It has shown several ways, and some of them do not translate.

Populations differ from patients. A highlander's body has been shaped over hundreds of generations to match its environment. A patient in shock has had minutes. Adaptation is not the same as resuscitation.

The observations are correlational. We see a low hemoglobin and a low rate of chronic mountain sickness, and we infer a link, but many other differences accompany the pattern, from diet to lung size to genetics elsewhere in the genome. The gene story is stronger than most, because the variant has been mapped and its effect on hemoglobin quantified, but even here the wider consequences are still being studied.

Finally, extrapolating from a healthy population to a sick one is a classic source of error. A lower hemoglobin is adaptive in a healthy Tibetan with intact vessels and lungs. It may be dangerous in a person with heart disease.

What I would carry into a design meeting

If I were sitting with a group of developers, I would bring four questions from the mountain and the sea.

Is your target patient limited by capacity, by flow, or by loading? A product that raises capacity does little for a patient whose problem is a blocked vessel, and a product that improves flow does little for a patient who is simply low on hemoglobin.

Where on the curve of viscosity and capacity does your patient sit? A patient already thin-blooded may gain little from added capacity, and one already thick-blooded, for example after a long stay at altitude or with a blood disorder, may be harmed by it.

What affinity does the environment call for? A single value for all uses is a compromise, and the compromise should be stated.

Are you transporting or storing? If the answer is storing, the product might look more like myoglobin than like blood.

The mountains do not answer those questions. They teach how to ask them.

Further reading

The comparison of hemoglobin concentration across highland populations is reviewed in a 2025 article in the American Journal of Human Biology on adaptive hemoglobin concentration at high altitude. The Tibetan EPAS1 variants were reported in a 2010 paper in the Proceedings of the National Academy of Sciences, and the Denisovan-like origin of the variant in a 2014 paper in Nature. Values quoted for disease prevalence come from the 2025 review and vary by study and definition. Statements about diving mammals reflect standard comparative physiology and should be confirmed in a specialist source before use.